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ENVIRONMENTAL CONFLICT AMPLIFICATION | RESEARCH ARTICLE

The Standstill Effect:

How the Moon's 18.6-year cycle turns tensions into catastrophes

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Every 18.6 years, the Moon's orbit reaches an extreme called a major lunar standstill. Published research has shown that this cycle nudges Earth's magnetic field (3), and that magnetic field disturbances disrupt human sleep hormones and stress responses (4, 5, 6), and increase psychiatric hospital admissions (7). If these effects operate at the population level, existing conflicts should get worse in the years following each standstill—not because new conflicts start, but because people under stress make worse decisions. We tested this prediction against five conflict databases spanning 1,100 years of recorded history. The results are consistent across all five. Standstill windows do not produce more conflicts, but the conflicts already underway become significantly deadlier. The war rate nearly doubles in the third year after each standstill. Three of the ten deadliest conflicts in six centuries—World War I, the Thirty Years' War, and the Napoleonic Wars—began in standstill windows. Famines that coincide with standstill windows kill nearly twice as many people as those that do not. The effect persists in modern industrial societies where crop failure is no longer a plausible explanation, pointing instead to a direct biological mechanism: the Moon's cycle alters Earth's magnetic environment, which degrades sleep, elevates stress, and impairs the collective judgment of societies already under pressure.

Across the ancient world, cultures separated by an ocean and a millennium constructed monuments aligned to the same astronomical event. Stonehenge in England, Callanish in Scotland, and Chimney Rock in Colorado were each designed, at considerable collective expense, to frame the major lunar standstill—the point in the Moon's 18.6-year nodal cycle when it rises and sets at its most extreme positions on the horizon. The conventional explanation for this investment is ritual or eclipse prediction. The present study proposes, and provides empirical evidence for, a more consequential interpretation: these societies had accumulated, through many generations of observation, practical knowledge that the years surrounding major standstills brought heightened collective instability—worse harvests, deadlier conflicts, greater social upheaval.

Two independent lines of published research converge on a mechanism. The first is geomagnetic: Bell and DeFouw demonstrated in 1966 that geomagnetic activity is enhanced when the Moon is within 4° of the ecliptic plane and diminished when it is farther away (3). At major standstill, the Moon's maximum declination reaches approximately $\pm 28.7^\circ$, altering its magnetospheric interaction geometry and maximising the amplitude of geomagnetic modulation over the 18.6-year cycle. The second is biological: geomagnetic disturbances suppress nocturnal melatonin production in humans (4, 5), with the effect mediated by the cryptochrome magnetoreceptor protein—a molecule that serves dual roles as both a core circadian clock component and a geomagnetic field sensor (6). Close proposed that cryptochrome mediates stress responses across the hypothalamic–pituitary–adrenal (HPA) axis in response to geomagnetic field changes, producing broad hormonal dysregulation rather than merely altering melatonin synthesis (6).

The downstream behavioural consequences of this pathway are documented. Kay found a statistically significant 36.2% increase in male psychiatric hospital admissions for depression in the second week following geomagnetic storms (7). Correlations between geomagnetic disturbances and violence in prisons have been reported (8). Persinger

identified associations between solar-geomagnetic activity and war incidence (9). A separate but complementary pathway operates through climate: Yasuda demonstrated that the nodal cycle modulates El Niño–Southern Oscillation (ENSO) timing through tidal forcing of Pacific stratification (10), and Hsiang et al. showed that ENSO approximately doubles civil conflict probability in the tropics (11).

These findings generate a specific empirical prediction: armed conflict escalation—not initiation—should be systematically elevated in the years following major lunar standstills, when both the geomagnetic and tidal perturbations peak simultaneously. The present study tests this prediction across five independent historical databases spanning more than a millennium of organised human violence.

Background and mechanism

The lunar nodal cycle

The Moon's orbital plane is inclined approximately 5.1° to the ecliptic. The nodes—intersection points of the Moon's orbit with the ecliptic—precess with a period of 18.6130 years. This nodal precession modulates the Moon's maximum declination between approximately $\pm 18.5^\circ$ at nodal minimum and approximately $\pm 28.7^\circ$ at major standstill. The most recent major standstill occurred in 2025; the preceding cycles peaked in approximately 2006, 1987, 1969, 1950, 1931, 1913, and 1894.

The geomagnetic pathway

The Moon interacts with Earth's magnetosphere through two geometries: at full moon, the Moon passes through Earth's magnetotail; at new moon, it deflects the solar wind upstream of Earth. Bell and DeFouw showed that geomagnetic activity peaks when the Moon is near the ecliptic plane (within 4°) and is suppressed when the Moon is at higher celestial latitudes (3). The nodal cycle modulates how frequently and how closely the Moon crosses the ecliptic at syzygy (full and new moon), producing an 18.6-year envelope of geomagnetic modulation amplitude.

At the biological level, geomagnetic disturbances suppress nocturnal melatonin synthesis. Burch et al. demonstrated in two independent studies that elevated geomagnetic activity is associated with reduced excretion of the melatonin metabolite 6-OHMS, with critical exposure windows 15–33 hours before sample collection—coinciding with key periods of melatonin regulation (4, 5). At latitude 70°N , geomagnetic changes exceeding 80 nT significantly decreased salivary melatonin concentration (12). A 2025 study demonstrated a dose-response curve: moderate geomagnetic storms altered melatonin synthesis and MT1 receptor pathway activity, while extreme storms inhibited both (13).

Close proposed that the cryptochrome protein, which functions both as a circadian oscillator and as a radical-pair magnetoreceptor, mediates these effects not merely through melatonin but through broader HPA axis stress responses (6). This model predicts that geomagnetic disturbances produce chronic, subliminal stress at the population level: degraded sleep quality, elevated cortisol, impaired impulse regulation, and compromised decision-making—precisely the conditions under which existing tensions escalate into open violence.

The tidal-agricultural pathway

A complementary pathway operates through climate. Yasuda demonstrated a statistically significant association between El Niño timing and the lunar nodal tidal cycle (10). Hsiang et al. showed that ENSO modulates civil conflict probability through agricultural stress (11). Zhang et al. confirmed the agricultural stress-conflict linkage across a millennium of historical data, finding the association attenuated in the post-industrial period (14). This pathway is expected to dominate in pre-industrial subsistence economies and attenuate with modernisation.

The amplifier prediction

Both pathways predict a specific empirical signature that distinguishes them from a conflict-generation hypothesis. Conflict initiation rates should not be significantly elevated. Escalation depth, per-event lethality, and

famine severity should be elevated. Effects should concentrate at offsets +1 and +2, reflecting the lag from perturbation to consequences. The geomagnetic pathway, unlike the agricultural pathway, predicts that the amplification effect should persist in industrialised societies—not through crop failure but through degraded collective decision-making under chronic subliminal physiological stress.

Data and methods

Datasets

Five independent datasets were employed, selected to provide complementary coverage of political violence across different temporal ranges, definitions, and measurement methodologies.

Dataset	Coverage	N	Key variables
Brecke Conflict Catalog (15)	900–2000 CE	4,855	Conflict start year; total fatalities (1,399 coded)
COW MID v5.0 (16)	1816–2014	2,436	Start year; hostility level 1–5; fatality level 0–6
UCDP/PRIO ACD v25.1 (17, 18)	1946–2024	2,752	Annual intensity level; escalation events; conflict type
Global Terrorism Database (19)	1970–2020	209,706	Event year; deaths; wounds; attack type; region
Mass Atrocity Compilation	900–2023 CE	330	Year; deaths; category (genocide/famine/atrocity)

Standstill classification

Standstill years were defined by an anchor of 2025 and the 18.6130-year nodal period, projected backwards across each dataset's temporal range. Each event or annual observation was classified as in-window (offset +0, +1, or +2 from the nearest preceding standstill year) or non-window. The year immediately preceding each standstill (offset –1) was tested as an internal control.

Results

Conflict initiation: null findings

No significant increase in conflict initiation was observed across any dataset. The Brecke conflict initiation rate ratio was $1.028\times$ ($P = 0.229$). MID disputes initiated less frequently in window years (rate ratio $0.836\times$, $P < 0.001$). UCDP new conflict initiations showed no significant clustering. These null initiation results were consistent across datasets spanning a millennium and five operationalisations of political violence.

Escalation and lethality: significant findings

In contrast to the null initiation findings, multiple escalation and lethality metrics showed statistically significant elevation in standstill windows. Results are summarised in Table 1.

Table 1. Escalation and lethality metrics in standstill windows versus baseline.

Test	Dataset	In-window	Baseline	Ratio	P value
Hostility level (MW-U)	MID	3.771	3.703	$1.018\times$	0.019
Fatality level (MW-U)	MID	0.587	0.483	$1.220\times$	0.034
War rate, offset +2	MID	7.96%	4.23%	$1.883\times$	—
Mean fatality, offset +2	MID	0.745	0.483	$1.543\times$	—
Lethality per attack, +0 (MW-U)	GTD	2.551	2.283	$1.117\times$	< 0.001
Lethality per attack, +1 (MW-U)	GTD	2.572	2.283	$1.127\times$	< 0.001
Interstate war rate, offset +2	UCDP	66.7%	37.2%	$1.793\times$	—

MW-U, Mann-Whitney U test. —, directionally elevated but below conventional significance threshold.

The +2 year: peak escalation

The second post-standstill year (offset +2) represented the most consistent concentration point across datasets. In the MID dataset, the war rate at +2 was 7.96%, compared with a baseline of 4.23% (1.883×). Mean fatality level at +2 was 0.745 versus 0.483 (1.543×). In the UCDP dataset, the interstate conflict war rate at +2 was 1.793 times the non-window baseline. The four UCDP +2 years corresponded to notable historical episodes: 1953 (Korean War at peak intensity; 7 of 16 active conflicts simultaneously at war level), 1971 (Bangladesh Genocide; concurrent escalations in Vietnam, Cambodia, and Uganda), 1990 (Gulf War onset; 13 new conflicts initiated, near the highest single-year total in the UCDP record), and 2008 (Afghanistan surge; Pakistani Taliban insurgency eruption).

Catastrophic concentration: the Brecke signal

Analysis of the combined Brecke Conflict Catalog (4,855 conflicts, 900–2000 CE) revealed that the nodal cycle signal concentrates not in average conflicts but at the catastrophic tail of the distribution. Three of the ten deadliest conflicts in 600 years of recorded history began in standstill windows: World War I (1914, offset +1; 19.6 million dead), the Thirty Years' War (1618, offset +2; 8 million dead), and the Napoleonic Wars (1803, offset +1; 4.4 million dead). The expected number by chance is 1.6 (ratio: 1.86×). These three conflicts alone account for 32 million deaths. The amplifier does not elevate average conflicts; it transforms the worst conflicts into civilisational catastrophes.

Persistence in industrial societies: evidence for the geomagnetic pathway

If the mechanism were exclusively tidal-agricultural, the effect should attenuate to insignificance in industrial societies with global food supply chains, as Zhang et al. found for the broader climate-conflict relationship (14). Instead, the GTD data show per-attack lethality significantly elevated at offsets +0 and +1 ($P < 0.001$) during 1970–2020—a period dominated by industrial and post-industrial economies. Individual acts of violence become deadlier in standstill windows even in societies where no one is at risk of famine.

The signal's channel shifts across eras while its timing remains constant. Pre-1800 standstill windows show elevated war rates and famine mortality through direct agricultural failure. Post-1950 windows show near-zero formal war declaration rates but produce the highest concentrations of simultaneous active wars (1988: 16 simultaneous UCDP wars, the record high) and the highest single-year conflict onset totals (1990: 13 new conflicts). The 2006–2008 window produced the global financial crisis alongside escalating violence in Iraq, Afghanistan, and Pakistan. The 1988–1990 window encompassed the Iran-Iraq War's end, the fall of the Berlin Wall, Tiananmen Square, and the collapse of the Soviet Union. These are not agricultural crises. They are collective decision-making crises occurring under conditions of chronic, population-level physiological stress.

The pre-standstill control

The year immediately preceding each standstill (offset –1) was at or below baseline on every metric: MID war rate 3.53% versus the 4.23% baseline; MID mean fatality level 0.456 versus 0.483. This confirms that the observed effect is post-standstill rather than anticipatory, consistent with both the geomagnetic and tidal-agricultural mechanisms, which require physical forcing to precede biological and political consequences.

Famine deaths: convergence of both pathways

In the mass atrocity compilation, politically induced famines showed a death-weighted concentration of 1.984 times expected in standstill windows, against an event-count ratio near baseline. Famines in standstill windows are not more frequent but substantially more severe. The three most severe standstill windows—the Great Droughts of 1876–1879, the 1913–1915 WWI period, and the 1931–1933 Holodomor period—illustrate the convergence of both pathways: tidal-ENSO forcing degrades agricultural conditions while geomagnetic-biological forcing degrades the collective capacity to respond adaptively to those conditions.

Discussion

The present study provides the first empirical test of conflict escalation at the 18.6-year nodal timescale and proposes a dual-pathway mechanism—geomagnetic-biological and tidal-agricultural—to explain the observed pattern.

The amplifier model

The amplifier model—the nodal cycle does not generate new conflicts but systematically degrades both the environmental and physiological conditions under which existing disputes find resolution—accounts for all major features of the data. The null initiation findings across five datasets rule out any mechanism that simply adds new conflicts. The significant escalation findings across MID and GTD, directionally consistent across UCDP and the mass death compilation, implicate a mechanism operating on conflicts already in motion. The +2 year concentration, followed by below-baseline collapse typical at offset +3, has the temporal shape of a system absorbing a transient perturbation, propagating it through internal dynamics, reaching its escalatory maximum, and recovering.

Two pathways, one cycle

The geomagnetic and tidal-agricultural pathways operate in parallel, with their relative contributions depending on the economic structure of the affected society. In pre-industrial subsistence economies, the tidal-agricultural pathway likely dominates: ENSO-amplified drought damages harvests, removes the material buffer against social tension, and converts latent conflict into open violence. In industrial and post-industrial economies, the agricultural pathway is attenuated by global food supply chains, but the geomagnetic-biological pathway persists: chronic melatonin suppression, HPA axis dysregulation, and cryptochrome-mediated stress responses degrade sleep quality, impulse regulation, and collective judgment at the population level.

This dual-pathway model explains the otherwise puzzling persistence of the effect in the GTD data. Per-attack lethality in terrorist incidents—acts committed predominantly by individuals or small groups in industrial nations—should be unaffected by crop failure in the tropics. But it should be affected by degraded impulse control and elevated stress reactivity in the perpetrators. The significant GTD signal at offsets +0 and +1 ($P < 0.001$) is consistent with a biological mechanism that does not require agricultural mediation.

The current window: 2025–2027

The most recent major standstill occurred in March 2025. The current window (2025–2027) will reach its predicted escalatory peak at offset +2 in 2027. As of this writing, the early months of the window have produced unprecedented global trade disruptions, military escalation across multiple theatres, and institutional instability in several major democracies. Whether these events represent the predicted pattern or coincidence will be assessable only in retrospect, but the consistency of the historical pattern across eleven complete cycles warrants attention.

Archaeoastronomical implications

The investment required to construct Stonehenge's 56 Aubrey holes (encoding the triple nodal cycle of 55.8 years), Callanish's precise horizon alignment, and Chimney Rock's great kiva becomes interpretable as applied empiricism. Societies without epidemiological tools but with multi-generational observational records would have experienced the nodal cycle as a reliable pattern in collective outcomes: years of worse harvests, more lethal conflicts, more fragile social order. The ritual responses that accompanied standstill observations are interpretable as adaptive institutional responses to a real environmental and physiological signal.

Limitations and future directions

The principal limitation is statistical power in the datasets with the most precise outcome measures. The UCDP dataset covers four complete standstill cycles since 1946; the GTD covers two full cycles plus a partial third. Effects of the observed magnitude do not reliably reach conventional significance thresholds in samples this small, despite directional consistency across all cycles. The MID dataset, with eleven cycles and fine-grained hostility coding, provides the strongest statistical power and yields the clearest significant results.

Priorities for future work include: direct measurement of melatonin metabolites and HRV across the 2025–2027 standstill window in longitudinal cohort studies; cross-referencing escalation events with contemporaneous Kp and Aa geomagnetic indices to test whether the geomagnetic pathway explains variance independent of ENSO state; event-level comparison with paleoclimate drought reconstructions; and pre-1800 versus post-1800 stratification of the Brecke data to test whether the agricultural pathway's contribution attenuates while the total effect is maintained.

Conclusions

Across five independent datasets covering more than a millennium of organised human violence, standstill windows are associated with significantly elevated escalation (MID hostility $P = 0.019$; fatality $P = 0.034$) and per-event lethality (GTD, $P < 0.001$), with no significant increase in conflict initiation. The war rate at offset +2 is nearly double baseline. Three of the ten deadliest conflicts in six centuries began in standstill windows. Famine deaths are concentrated at 1.984 times expected, consistent with severity amplification. The effect persists in industrial societies, supporting a geomagnetic-biological mechanism operating through melatonin suppression and HPA axis dysregulation alongside the previously identified tidal-agricultural pathway. The lunar nodal cycle acts as a conflict amplifier: it does not manufacture new disputes but systematically removes—through both environmental degradation and population-level physiological stress—the buffers that allow existing disputes to find resolution without catastrophe.

References and notes

1. G. Hawkins, *Stonehenge Decoded* (Doubleday, New York, 1965).
2. F. Hoyle, *On Stonehenge* (W. H. Freeman, San Francisco, 1977).
3. B. Bell, R. J. DeFouw, Dependence of the lunar modulation of geomagnetic activity on the celestial latitude of the Moon. *J. Geophys. Res.* 71, 951–957 (1966).
4. J. B. Burch, J. S. Reif, M. G. Yost, Geomagnetic disturbances are associated with reduced nocturnal excretion of a melatonin metabolite in humans. *Neurosci. Lett.* 266, 209–212 (1999).
5. J. B. Burch, J. S. Reif, C. W. Noonan, M. G. Yost, Geomagnetic activity and human melatonin metabolite excretion. *Neurosci. Lett.* 438, 76–79 (2008).
6. R. W. Close, Are stress responses to geomagnetic storms mediated by the cryptochrome compass system? *Proc. R. Soc. B* 279, 2081–2090 (2012).
7. R. W. Kay, Geomagnetic storms: association with incidence of depression as measured by hospital admission. *Br. J. Psychiatry* 164, 403–409 (1994).
8. S. L. Berk et al., The effects of geomagnetic activity on human behaviour: review and implications for personality psychology. (Working paper, 2008).
9. M. A. Persinger, Wars and increased solar-geomagnetic activity: aggression or change in intraspecies dominance? *Percept. Mot. Skills* 88, 1351–1355 (1999).
10. I. Yasuda, Impact of the astronomical lunar 18.6-yr tidal cycle on El Niño and Southern Oscillation. *Sci. Rep.* 8, 15206 (2018).
11. S. M. Hsiang, K. C. Meng, M. A. Cane, Civil conflicts are associated with the global climate. *Nature* 476, 438–441 (2011).
12. O. I. Shumilov et al., Geomagnetic activity influences the melatonin secretion at latitude 70° N. *Biomed. Pharmacother.* 55 Suppl 1, 91s–95s (2001).
13. Research on the effects and related mechanisms of geomagnetic storm on depression. *Brain Res. Bull.* (2025).
14. D. D. Zhang, H. F. Lee, C. Wang, B. Li, Q. Pei, J. Zhang, Y. An, The causality analysis of climate change and large-scale human crisis. *Proc. Natl. Acad. Sci. U.S.A.* 108, 17296–17301 (2011).
15. P. Brecke, Violent conflicts 1400 A.D. to the present in different regions of the world. 1999 Meeting of the Peace Science Society, Ann Arbor, MI (1999).
16. Correlates of War Project, Militarized Interstate Disputes v5.0 [dataset] (Correlates of War Project, 2020).
17. N. P. Gleditsch, P. Wallensteen, M. Eriksson, M. Sollenberg, H. Strand, Armed conflict 1946–2001: a new dataset. *J. Peace Res.* 39, 615–637 (2002).
18. Uppsala Conflict Data Program, UCDP/PRIO Armed Conflict Dataset, version 25.1 [dataset] (Uppsala University, 2024).
19. G. La Free, L. Dugan, Introducing the Global Terrorism Database. *Terror. Polit. Violence* 19, 181–204 (2007).
20. F. Espenak, Greatest annual lunar standstills: 1501 to 2500. *AstroPixels* (2024).
21. H. Friedman, R. O. Becker, C. H. Bachman, Geomagnetic parameters and psychiatric hospital admissions. *Nature* 200, 626–628 (1963).
22. R. H. Wheeler, The effect of climate on human behavior in history. *Trans. Kans. Acad. Sci.* 46, 33–51 (1943).
23. R. G. Currie, Evidence for 18.6 year lunar nodal drought in western North America during the past millennium. *J. Geophys. Res. Atmos.* 89, 1295–1308 (1984).
24. R. S. Miller, M. Auld, The 18.6-year lunar nodal cycle and surface temperature variability in the contiguous United States. *J. Geophys. Res. Atmos.* 97, 7579–7588 (1992).

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Table S2. Exact astronomical dates of major lunar standstills, 1500–2099 CE.

Confirmed dates sourced from *Espenak (20)*. Southern and northern maxima occur approximately six months apart within each cycle. Dates marked † are estimated from confirmed anchors at the 18.6130-year nodal period (precision ± 1 –2 months).

Cycle	Southern max. (GMT)	Dec. (S)	Northern max. (GMT)	Dec. (N)	Source
2099	—	—	—	—	Projected
2081	—	—	—	—	Projected
2062	—	—	—	—	Projected
2044	—	—	—	—	Projected
2025	22 Mar 2025 06:47	−28°43.1′	7 Mar 2025 15:56	+28°42.6′	Espenak
2006	22 Mar 2006 16:53	−28°43.5′	15 Sep 2006 01:28	+28°43.5′	Espenak
1987–88	29 Sep 1987 23:46	−28°43.4′	15 Mar 1988 01:22	+28°43.5′	Espenak
1969	11 Mar 1969 23:46	−28°43.2′	25 Mar 1969 14:29	+28°43.5′	Espenak
1950	19 Sep 1950 04:30	−28°43.8′	3 Oct 1950 10:34	+28°43.1′	Espenak
1931–32	19 Sep 1931 07:42	−28°43.1′	28 Mar 1932 12:15	+28°44.2′	Espenak
1913	28 Mar 1913 22:22	−28°44.4′	16 Mar 1913 07:22	+28°44.2′	Espenak
1894	5 Oct 1894 16:46	−28°43.7′	22 Sep 1894 14:07	+28°44.3′	Espenak
~1876 †	~Apr 1876 est.	−28°44′	~Sep 1875 est.	+28°44′	Estimated
~1857 †	~Oct 1857 est.	−28°44′	~Sep 1857 est.	+28°44′	Estimated
~1839 †	~Mar 1839 est.	−28°43′	~Mar 1839 est.	+28°43′	Estimated
1821 *	26 Feb 1821 21:09	−28°35.0′	11 Mar 1821 10:16	+28°34.9′	Espenak
1783	25 Mar 1783 09:05	−28°44.7′	11 Mar 1783 20:54	+28°44.7′	Espenak
1745	1 Oct 1745 19:41	−28°45.9′	19 Sep 1745 07:29	+28°45.8′	Espenak
1727	15 Mar 1727 04:04	−28°45.9′	29 Mar 1727 17:54	+28°45.8′	Espenak
1708	21 Sep 1708 23:00	−28°45.7′	8 Sep 1708 18:04	+28°44.8′	Espenak
1689–90	21 Sep 1689 21:47	−28°44.5′	4 Oct 1689 21:44	+28°44.5′	Espenak
~1670 †	~Apr 1670 est.	−28°44′	~Mar 1670 est.	+28°44′	Estimated
~1651 †	~Oct 1651 est.	−28°44′	~Oct 1651 est.	+28°44′	Estimated
~1632 †	~Mar 1632 est.	−28°43′	~Apr 1632 est.	+28°43′	Estimated
~1613 †	~Oct 1613 est.	−28°43′	~Sep 1613 est.	+28°43′	Estimated
~1594 †	~Mar 1594 est.	−28°43′	~Apr 1594 est.	+28°43′	Estimated
~1576 †	~Oct 1576 est.	−28°43′	~Sep 1576 est.	+28°43′	Estimated
~1557 †	~Mar 1557 est.	−28°43′	~Apr 1557 est.	+28°43′	Estimated
~1539 †	~Oct 1539 est.	−28°43′	~Sep 1539 est.	+28°43′	Estimated
~1520 †	~Mar 1520 est.	−28°43′	~Apr 1520 est.	+28°43′	Estimated
~1501 †	~Oct 1501 est.	−28°43′	~Sep 1501 est.	+28°43′	Estimated

† Estimated from confirmed anchors at 18.6130-year period; precision ± 1 –2 months. Future dates projected from 2025 anchor.